

Maintenance Wizard for Steel Plant - Simple Explanation

1. Problem

In steel plants, engineers must check manuals, SOPs, sensor data, logs, and maintenance records from different places. This takes time, increases downtime, and depends heavily on expert knowledge.

2. Solution

Build an AI-powered Maintenance Wizard that acts like ChatGPT for maintenance engineers. It analyzes machine data, manuals, SOPs, and historical records to provide recommendations.

3. Inputs

Equipment logs, error messages, failure reports, sensor data, anomaly alerts, manuals, SOPs, maintenance history, spare-part data, and engineer questions.

4. Outputs

Fault diagnosis, root-cause analysis, risk levels, maintenance actions, spare recommendations, failure predictions, alerts, and reports.

5. RAG Knowledge System

Manuals and SOPs are stored in a vector database. When an engineer asks a question, relevant information is retrieved and provided to the LLM for accurate answers.

6. Sensor Monitoring

Temperature, vibration, pressure, and current sensors continuously send data. The system detects abnormalities and generates alerts.

7. Failure Prediction

Historical failures and sensor trends are analyzed to estimate equipment health and Remaining Useful Life (RUL).

8. Risk Prioritization

Machines are ranked as Critical, High, Medium, or Low priority based on business impact, risk, and spare availability.

9. Dashboard

Shows plant health, machine status, active alerts, predicted failures, and spare shortages.

10. User Journey

Engineer opens dashboard -> sees alert -> asks AI -> gets diagnosis -> receives maintenance plan -> generates report.

11. Features Judges Will Like

AI Chat Assistant, RAG-based knowledge retrieval, failure prediction, risk scoring, maintenance reports, and real-time alerts.

12. Expected Outcome

Reduced downtime, faster troubleshooting, better maintenance planning, improved reliability, and proactive maintenance.